2014 PJC J2 Prelim Examination H2 Maths Paper 2 Solution

1 (i) $$\frac{6n+2}{(n+1)(n+2)(n+3)}=\frac{A}{n+1}+\frac{B}{n+2}+\frac{C}{n+3}$$

$$6n+2=A(n+2)(n+3)+B(n+1)(n+3)+C(n+1)(n+2)$$
$$n=-1: \quad A=-2$$
$$n=-2: \quad B=10$$
$$n=-3: \quad C=-8$$
$$\therefore \frac{6n+2}{(n+1)(n+2)(n+3)}=-\frac{2}{n+1}+\frac{10}{n+2}-\frac{8}{n+3}$$

(ii) $$\sum_{r=2}^{n}\left(\frac{3r+1}{(r+1)(r+2)(r+3)}\right)$$

$$=\frac{1}{2}\sum_{r=2}^{n}\left(-\frac{2}{r+1}+\frac{10}{r+2}-\frac{8}{r+3}\right)$$

$$\begin{aligned}
= \quad & \left(-\frac{1}{3}+\frac{5}{4}-\cancel{\frac{4}{5}}\right) \\
&+\left(-\frac{1}{4}+\cancel{\frac{5}{5}}-\cancel{\frac{4}{6}}\right) \\
&+\left(-\cancel{\frac{1}{5}}+\cancel{\frac{5}{6}}-\cancel{\frac{4}{7}}\right) \\
&+\left(-\cancel{\frac{1}{6}}+\cancel{\frac{5}{7}}-\cancel{\frac{4}{8}}\right) \\
&\qquad \vdots \\
&+\left(-\cancel{\frac{1}{n-1}}+\cancel{\frac{5}{n}}-\cancel{\frac{4}{n+1}}\right) \\
&+\left(-\cancel{\frac{1}{n}}+\cancel{\frac{5}{n+1}}-\frac{4}{n+2}\right) \\
&+\left(-\cancel{\frac{1}{n+1}}+\frac{5}{n+2}-\frac{4}{n+3}\right)
\end{aligned}$$

$$=-\frac{1}{3}+\frac{5}{4}-\frac{1}{4}-\frac{4}{n+2}+\frac{5}{n+2}-\frac{4}{n+3}$$
$$=\frac{2}{3}+\frac{1}{n+2}-\frac{4}{n+3}$$

(iii)

$$\sum_{r=2}^{n}\left(\frac{3r+1}{(r+1)(r+2)(r+3)}\right)=\frac{7}{3(4)(5)}+\frac{10}{4(5)(6)}+\frac{13}{5(6)(7)}+\cdots+\frac{3n-2}{n(n+1)(n+2)}+\frac{3n+1}{(n+1)(n+2)(n+3)}$$

$$\sum_{k=2}^{n}\left(\frac{3k-2}{k(k+1)(k+2)}\right)=\frac{4}{2(3)(4)}+\frac{7}{3(4)(5)}+\frac{10}{4(5)(6)}+\frac{13}{5(6)(7)}+\cdots+\frac{3n-2}{n(n+1)(n+2)}$$

$$=\frac{4}{2(3)(4)}+\sum_{k=2}^{n-1}\left(\frac{3k+1}{(k+1)(k+2)(k+3)}\right)$$

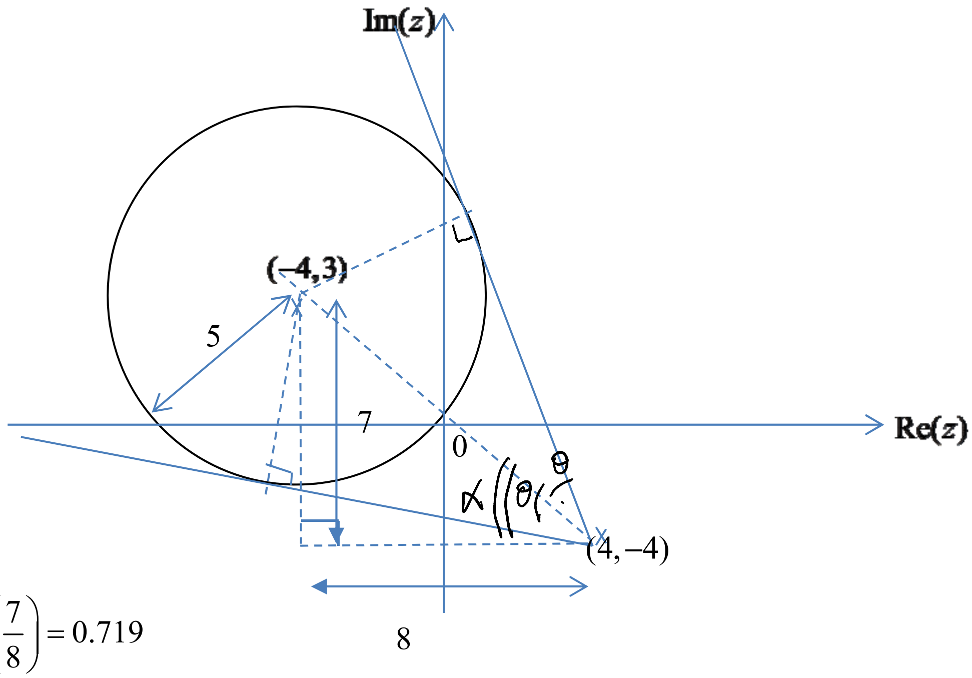
$$=\frac{1}{6}+\frac{2}{3}+\frac{1}{n+1}-\frac{4}{n+2}=\frac{5}{6}+\frac{1}{n+1}-\frac{4}{n+2}$$

2

$|6i-8-2z|=|6-8i|$

$\Rightarrow |-2||z+4-3i|=10$

$\Rightarrow |z-(-4+3i)|=5$



$$\alpha=\tan^{-1}\left(\frac{7}{8}\right)=0.719$$

$$\theta=\sin^{-1}\left(\frac{5}{\sqrt{(4+4)^2+(-4-3)^2}}\right)=\sin^{-1}\left(\frac{5}{\sqrt{113}}\right)=0.490$$

Smallest $\arg(z+4i-4)=\pi-0.719-0.490=1.93\,\text{rad}$

Largest $\arg(z+4i-4)=\pi-0.719+0.490=2.91\,\text{rad}$

3 (i)

$$\int x^2 e^x \, dx = x^2 e^x - 2\int x e^x dx + C'$$

$$= x^2 e^x - 2\left(x e^x - \int e^x dx\right) + C'$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

$$= e^x\left(x^2 - 2x + 2\right) + C$$

$u = x^2$ $\quad \frac{dv}{dx} = e^x$

$\frac{du}{dx} = 2x$ $\quad v = e^x$

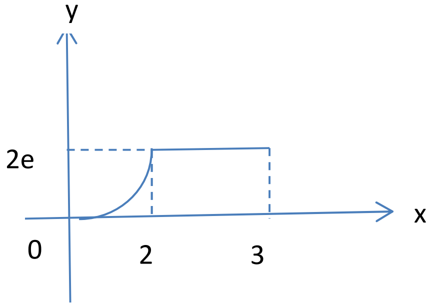
$u = x$ $\quad \frac{dv}{dx} = e^x$

$\frac{du}{dx} = 1$ $\quad v = e^x$

(ii)

$y = 2e \quad \Rightarrow x e^{\frac{1}{2}x} = 2e$

$\Rightarrow x = 2 \ (\text{by observation})$



Required volume

$$= \pi \int_0^2 \left(x e^{\frac{1}{2}x}\right)^2 dx + \pi(2e)^2$$

$$= \pi \int_0^2 x^2 e^x dx + 4\pi e^2$$

$$= \pi\left[e^x\left(x^2 - 2x + 2\right)\right]_0^2 + 4\pi e^2$$

$$= \pi\left(e^2(4 - 4 + 2) - 2\right) + 4\pi e^2$$

$$= 6\pi e^2 - 2\pi$$

4 (i)

$\sin\sqrt{y} = x \quad \Rightarrow y = \left(\sin^{-1} x\right)^2$

$$\Rightarrow \frac{dy}{dx} = 2\left(\sin^{-1} x\right) \times \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{4y}{1-x^2}$$

$$\Rightarrow \left(1-x^2\right)\left(\frac{dy}{dx}\right)^2 = 4y$$

$$\Rightarrow -2x\left(\frac{dy}{dx}\right)^2 + \left(1-x^2\right)2\frac{dy}{dx}\frac{d^2y}{dx^2} = 4\frac{dy}{dx} \qquad \Rightarrow \left(1-x^2\right)\frac{d^2y}{dx^2} = x\frac{dy}{dx} + 2 \qquad \text{(proved)}$$

(ii)

$$\left(1-x^2\right)\frac{d^2y}{dx^2}=x\frac{dy}{dx}+2$$

$$\Rightarrow -2x\frac{d^2y}{dx^2}+\left(1-x^2\right)\frac{d^3y}{dx^3}=\frac{dy}{dx}+x\frac{d^2y}{dx^2}$$

$$\Rightarrow \left(1-x^2\right)\frac{d^3y}{dx^3}=3x\frac{d^2y}{dx^2}+\frac{dy}{dx}$$

$$\Rightarrow -2x\frac{d^3y}{dx^3}+\left(1-x^2\right)\frac{d^4y}{dx^4}=3\frac{d^2y}{dx^2}+3x\frac{d^3y}{dx^3}+\frac{d^2y}{dx^2}$$

$$\Rightarrow \left(1-x^2\right)\frac{d^4y}{dx^4}=5x\frac{d^3y}{dx^3}+4\frac{d^2y}{dx^2}$$

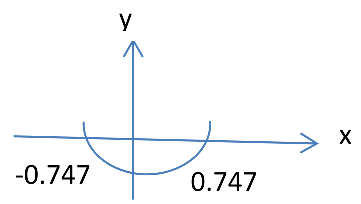
$$x=0\Rightarrow y=0,\frac{dy}{dx}=0,\frac{d^2y}{dx^2}=2,\frac{d^3y}{dx^3}=0,\frac{d^4y}{dx^4}=8\Rightarrow y\approx x^2+x^4\frac{8}{4!}=x^2+\frac{1}{3}x^4$$

(iii)

$$y=\left(\sin^{-1}x\right)^2, Y=x^2+\frac{1}{3}x^4$$

$$\left|y-Y\right|<0.05$$

By GC, $\quad -0.747<x<0.747$



5

(i) Since N is on l_1,

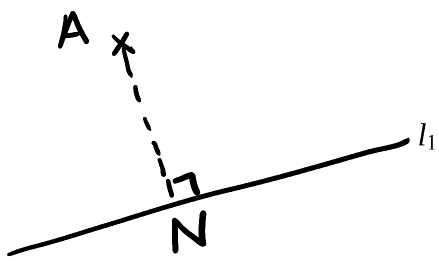
$\overrightarrow{ON}=\begin{pmatrix}-3+2\lambda\\10-4\lambda\\-5+\lambda\end{pmatrix}$ for some value of λ.

$$\overrightarrow{AN}=\begin{pmatrix}-3+2\lambda\\10-4\lambda\\-5+\lambda\end{pmatrix}-\begin{pmatrix}-1\\-2\\6\end{pmatrix}=\begin{pmatrix}-2+2\lambda\\12-4\lambda\\-11+\lambda\end{pmatrix}$$

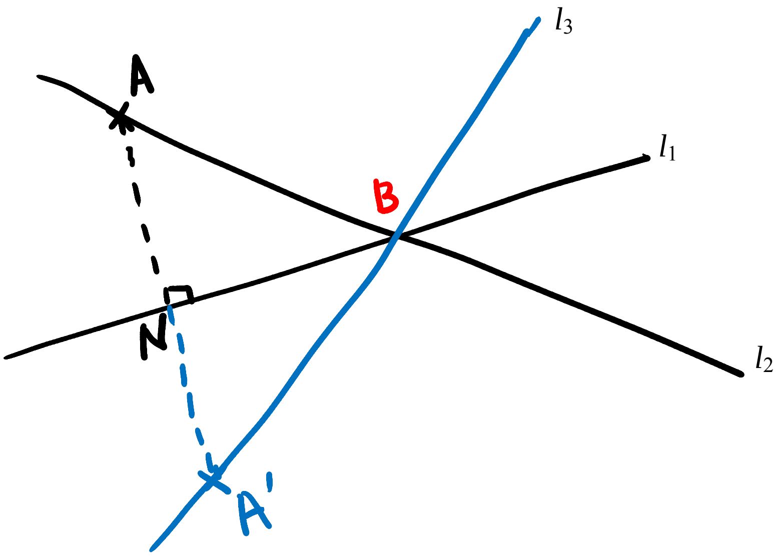
Since $\overrightarrow{AN}\perp$ to l_1, $\overrightarrow{AN}\bullet\begin{pmatrix}2\\-4\\1\end{pmatrix}=0$

$$\therefore \begin{pmatrix}-2+2\lambda\\12-4\lambda\\-11+\lambda\end{pmatrix}\bullet\begin{pmatrix}2\\-4\\1\end{pmatrix}=0\Rightarrow \lambda=3$$

Therefore, $\overrightarrow{ON}=\begin{pmatrix}3\\-2\\-2\end{pmatrix}$



(ii) Since $(-3,10,-5)$ lies on both lines, $(-3,10,-5)$ is the intersection point of l_1 and l_2



Let point A' be the reflection of point A about l_1 and point B be the intersection between l_1 and l_2

Note:

You may also use

$$\overrightarrow{BN}=\frac{\overrightarrow{BA}+\overrightarrow{BA'}}{2}$$

Using ratio theorem, $\overrightarrow{ON}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$

$$\Rightarrow \overrightarrow{OA'}=2\overrightarrow{ON}-\overrightarrow{OA}\Rightarrow \overrightarrow{OA'}=2\begin{pmatrix}3\\-2\\-2\end{pmatrix}-\begin{pmatrix}-1\\-2\\6\end{pmatrix}=\begin{pmatrix}7\\-2\\-10\end{pmatrix}$$

$$\overrightarrow{A'B}=\begin{pmatrix}-3\\10\\-5\end{pmatrix}-\begin{pmatrix}7\\-2\\-10\end{pmatrix}=\begin{pmatrix}-10\\12\\5\end{pmatrix}$$

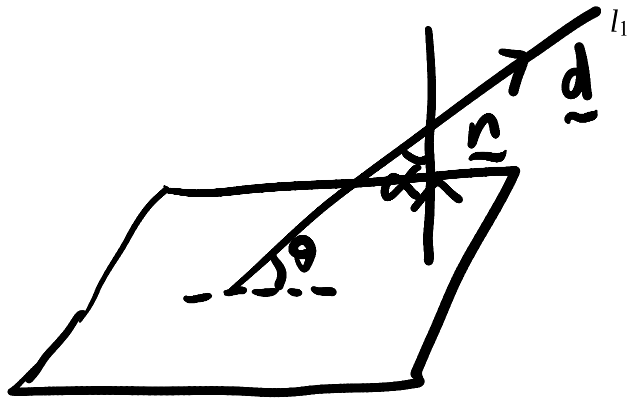
Therefore,

Vector equation of line of reflection of the line l_2 in l_1 is

$$\mathbf{r}=\begin{pmatrix}7\\-2\\-10\end{pmatrix}+\alpha\begin{pmatrix}-10\\12\\5\end{pmatrix}\Rightarrow\frac{x-7}{-10}=\frac{y+2}{12}=\frac{z+10}{5}$$ or

$$\mathbf{r}=\begin{pmatrix}-3\\10\\-5\end{pmatrix}+\alpha\begin{pmatrix}-10\\12\\5\end{pmatrix}\Rightarrow\frac{x+3}{-10}=\frac{y-10}{12}=\frac{z+5}{5}$$

(iii)



Since p is perpendicular to l_2, Normal of $p = \begin{pmatrix} -2 \\ 12 \\ -11 \end{pmatrix}$

$$\left| \begin{pmatrix} -2 \\ 12 \\ -11 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} \right| = \sqrt{(-2)^2 + 12^2 + (-11)^2} \sqrt{2^2 + (-4)^2 + 1^2} \cos\alpha$$

$\Rightarrow 63 = \sqrt{269}\sqrt{21}\cos\alpha$

$\Rightarrow \alpha = 33.048°$

Therefore, $\theta = 57.0°$

6 Use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A \cap B) = P(A)P(B)$ if A and B indept

$$\frac{1}{2} = k + \frac{1}{6} + k + \frac{1}{12} - (k + \frac{1}{6})(k + \frac{1}{12})$$

$$\frac{1}{2} = 2k + \frac{3}{12} - k^2 - \frac{3}{12}k - \frac{1}{72}$$

$$k^2 - \frac{7}{4}k + \frac{19}{72} = 0 \quad \text{or} \qquad 72k^2 - 126k + 19 = 0$$

$$k = \frac{1}{6} \text{ or } \frac{19}{12} \text{ (NA)}$$

7 (i) Prob required = $\dfrac{{}^3C_1\,{}^{17}C_2}{{}^{20}C_3}$ =0.358

(ii) Prob required = $\dfrac{{}^3C_1\,{}^6C_1\,{}^7C_1 + {}^3C_1\,{}^6C_1\,{}^4C_1 + {}^3C_1\,{}^7C_1\,{}^4C_1 + {}^6C_1\,{}^7C_1\,{}^4C_1}{{}^{20}C_3}$ =0.395

(iii) Prob required = P(two male from Production Dept) + P(3 male from Production Dept)

$$= \frac{{}^5C_2\,{}^{15}C_1 + {}^5C_3}{{}^{20}C_3} = \frac{160}{1140} = 0.140$$

Alternatively Prob required = 1 – P(no male from Production Dept) – P(1 male from Production dept)

$$= 1 - \frac{{}^{5}C_1\,{}^{15}C_2 + {}^{5}C_0}{{}^{20}C_3} = 0.140$$

8 (i) From GC, $\bar{x} = 50.631 \approx 50.6$

$s = 1.25755 \approx 1.26$

Let μ be the mean breaking strengths of fibre

$H_0: \mu = 50$ $H_1: \mu > 50$

Test Statistic T = $\dfrac{\bar{x}-\mu}{s/\sqrt{n}}$ =1.58673

p-value = 0.0735 > 0.04

So do not reject null hypothesis and conclude that there is insufficient evident to support company claim at 4% significance level.

Assume that the breaking strength of fibre follows normal distribution

There is a probability of 0.04 of wrongly conclude that the breaking strength is more than 50 netwon when it is actually equal to 50 newton.

(ii) $H_0: \mu = 50$ $H_1: \mu > 50$

Use of z-test, Test Statistic Z = $\dfrac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ =1.629

p-value = 0.0516

In order to support the company claim, we need to reject the null hypothesis.

p-value < $\dfrac{\alpha}{100}$, so α > 5.16

9 (i) (a) The probability of willing to take part is 0.3 remains constant for all patients

(b) All patients willing to take parts are independent

(i) The patients may not be independent as the disease is genetic, may affect same people from same family

(ii) R ~ number of patients willing to take part in the trial out of 8 patients

$R \sim B(8, 0.3)$

$P(R \geq 2) = 1 - P(R \leq 1) = 0.7447 \approx 0.745$

(iii) R ~ number of patient willing to take part in the trial out of n patients

$R \sim B(n, 0.3)$

$P(R \geq 2) > 0.9$

$1 - P(R \leq 1) > 0.9$

P($R \leq 1$) < 0.1
From GC:
$n = 11$, P($R \leq 1$) = 0.11299
$n = 12$, P($R \leq 1$) = 0.08503
$n = 13$, P($R \leq 1$) = 0.06367
So $n = 12$

(iv) As she asks patients who come to her clinic suffering from this disease on a particular day, patient not coming on that day is not selected ,each patient is not selected with equal chance, so it is not a random sample.

Decide a quota of 50 based on age groups, say 20-30, 31-40, 41-50, >50. Eg:

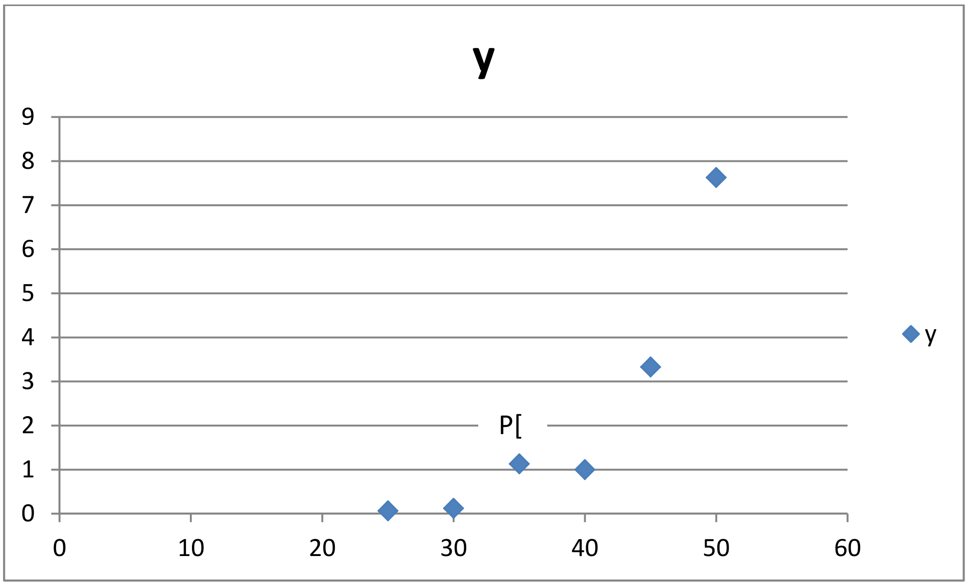
Age group	20-30	31-40	41-50	>50
Number	10	10	10	20

Ask the patients with the disease until the quota is met in each age group.

10 (i) The scatter diagram provides the information whether the points are linearly correlated. We could have a sample of data follows a curve but with linear correlation close to 1 or -1.

(ii) $r = 0.870$, indicates a strong positive linear correlation between x and y, so linear regression is appropriate.

(iii)



(iv) Once the point P is removed, the remaining points follows a curve that increases in a manner that the increment of y increases when x increases, which suggests that the data points may fit better with $\ln y = a + bx$.

(v) The new table is as follow:

x	25	30	40	45	50
$\ln y$	–2.703	–2.079	0	1.203	2.0317

From GC, $\ln y = -7.7838 + 0.19669x$

(vi) $x = 35$, $\ln y = -7.7838 + 0.19669\,(35) = -0.89965$

$y = 0.407$

(vii) Since x is independent and y is dependent variables, both equations cannot be used to find x.

Remarks:

Point 3 is the outliner as hint in (vii) mention it. In the marking, we also accept Point 4 as the outliner, answer in (v) will be $\ln y = -7.40808 + 0.192517x$ and (vi) $y = 1.34$

11

$$\mathrm{P}(X < 800) = 0.023$$

$$\frac{800-\mu}{\sigma} = -1.99539$$

$\mu - 1.99539\sigma = 800$ ----- ------(1)

$$\mathrm{P}(X > 1100) = 0.281$$

$$P(X \le 1100) = 0.719$$

$$\frac{1100-\mu}{\sigma} = 0.57987$$

$$\mu + 0.57987\sigma = 1100 - - - - - (2)$$

Solving, $\mu = 1032.45$ and $\sigma = 116.49$

(i) Y~number of batteries have lifetime of less than 800 hours, out of 200

Y~ B(200, 0.023)

*Since n is large, np = 4.6 < 5, so Y~*Po(4.6) approx.

$\mathrm{P}(Y \le 10) = 0.992$

(ii) $\overline{X} \sim N(1032.45, \frac{116.49^2}{n})$

$P(\overline{X} \leq 1000) > 0.1$

From GC, n = 20, $P(\overline{X} \leq 1000) = 0.10642$

n = 21, $P(\overline{X} \leq 1000) = 0.10088$

n=22, $P(\overline{X} \leq 1000) = 0.09568$

So largest n = 21

Alternatively

$$P\left(Z \leq \frac{1000-1032.45}{116.49/\sqrt{n}}\right) > 0.1$$

$$-\frac{32.45\sqrt{n}}{116.49} > -1.28155$$

$$\sqrt{n} < 4.60055$$

$$n < 21.165$$

So largest n = 21

12 Let $U \sim$ number of breakdown in A in a week, $U \sim Po(3.5)$

$V \sim$ number of breakdown in B in a week, $V \sim Po(2.8)$

$U + V \sim Po(6.3)$

$P(U + V < 4) = P(U + V \leq 3) = 0.12637 \approx 0.126$

(ii) Probability required = $\frac{P(U > V \text{ and } U + V < 4)}{P(U + V < 4)}$

$$= \frac{P(U=1)P(V=0) + P(U=2)P(V \leq 1) + P(U=3)P(V=0)}{P(U+V<4)}$$

$= 0.49290 \approx 0.493$

(iii) Let $X \sim$ number of breakdowns in A computer in 100 days, $X \sim Po(50)$

$Y \sim$ number of breakdowns in B computer in 100 days, $Y \sim Po(40)$

Since both $\lambda > 10$, $X \sim N(50, 50)$ approx and $Y \sim N(40, 40)$ approx

$X - 2Y \sim N(-30, 210)$

$P(|X - 2Y| > 5) = 1 - P(-5 \leq X - 2Y \leq 5) = 1 - P(-5.5 < X - 2Y < 5.5)$ (c.c.)

$= 0.962$

(iv) The number of breakdown in A is independent to that in B.